

# Dexter Fernandes

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https://github.com/Dexter-Fernandes

## AI Engineer

AI engineer with 3+ years building production ML systems in Python, now focused on RAG pipelines and LLM evaluation. Built a retrieval pipeline end to end (recursive chunking, embeddings, a FAISS vector store, top-k retrieval, generation) with an LLM-as-judge harness that separates retrieval failures from generation failures. Strongest on evaluation, observability, and guardrails: schema-validated judge contracts, reproducible config-driven runs measured against a baseline. Earlier work centred on deployment reliability (structured logging, health checks).

## PROJECTS

### FinanceBench RAG Evaluation Harness

Jun 2026

- Built an end-to-end dense-retrieval RAG pipeline for financial-document QA: PDF pages to recursive text chunking, embeddings, FAISS vector store, top-k retrieval, and LLM generation.
- Implemented three comparable retrieval modes (closed-book, oracle-context, and dense retrieval) so retrieval quality could be isolated from generation quality by ablating the context source.
- Built a dual-track scoring system pairing deterministic checks (exact/normalised match, numeric extraction) with an LLM-as-judge, grading answers on both hard correctness and qualitative grounding.
- Engineered scale-aware financial value matching that normalises "\$1.5B" against "1,500 million" against "1500000000" with tolerance-based comparison, handling the unit/scale ambiguity that trips naive string-match evaluators on finance data.
- Designed the LLM judge with schema-validated output contracts and a typed error path that logs malformed model output instead of silently corrupting scores, an observability guardrail for non-deterministic evaluators.
- Built retrieval-vs-answer metric joining to attribute each wrong answer to a retrieval miss, a generation error, or a grounding failure, with Markdown/CSV/JSON reports and a compare-runs mode for A/B experiments measured against a baseline.

## WORK EXPERIENCE

### Independent Projects & Upskilling

Sep 2024 - Present

#### Self-directed

- Built an end-to-end RAG evaluation harness for financial-document QA, covering both the retrieval pipeline and an LLM-as-judge scoring layer (full detail in Projects).
- Completed Nvidia "Building RAG Agents with LLMs" (Jul 2026), covering RAG, agents, and evaluation.

### Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

#### AI Engineer

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on Bristish Petroleum rigs.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Led on-site validation and diagnostics on deployments, troubleshooting model failures under environmental stressors and translating performance findings into fixes that improved robustness in live operation.

- Hardened deployed AI services with Dockerised microservices, Kafka messaging, structured logging, health checks, retries, and safe restarts reducing live video pipeline crashes by 67% and improved root-cause analysis.
- Presented model performance, failure modes, and deployment trade-offs to engineering teams, client stakeholders, and field operators, turning messy production signals into clear optimisation recommendations.

**LivNSense Technologies • Bengaluru, India**

**May 2020 - Dec 2022**

**Lead Software Engineer - AI**

- Led a team to design, train, and deploy production object detection models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries and John Deere.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using Python and OpenCV.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% on large real-world datasets.
- Ran systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Used confusion-matrix failure clustering to diagnose recurring models failure, to prioritise dataset fixes, and improve model robustness across camera feeds.
- Worked with hardware, software, QA, and client teams to translate model performance, latency, and failure analysis into rollout decisions for deployed safety systems.

## EDUCATION

**Master of Science in Robotics - Merit**

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

**Dissertation:** Radiological SLAM with LiDAR Odometry

- Dissertation: designed and implemented a Radiological SLAM system on a Clearpath Husky platform, integrating LiDAR and radiological data for localisation, mapping, and inspection workflows with Python.

**Bachelor of Technology in Electrical and Electronics Engineering**

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

## CERTIFICATIONS

**Building RAG Agents with LLMs**

Nvidia

Jul 2026

**Deep Learning Specialisation**

deeplearning.ai

Jul 2019

## SKILLS

- **AI Engineering and Evaluation:** LLM-as-judge, RAG evaluation, document-grounded QA, retrieval metrics, evidence matching, citation checks, hallucination detection, error analysis, structured outputs, RAG pipelines, dense retrieval, FAISS vector store, ChromaDB
- **Experimentation and Reliability:** failure-mode analysis, benchmarking, regression checks, hypothesis testing, ablation studies, post-deployment validation, production monitoring
- **Python and ML Engineering:** Python, Pandas, NumPy, scikit-learn, PyTorch, MLflow, DVC, Apache Airflow, Optuna, LangChain, LangSmith
- **Production Systems:** Docker, CI/CD, Git, structured logging, observability, health checks, Apache Kafka, Linux, schema-validated output contracts, Microsoft Azure